# Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

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***Abstract*—On-track system identification promises tire models learned from ordinary racing laps instead of dedicated steady-state manoeuvres. We study one such residual-learning pipeline at full vehicle scale, in simulation, on a plant that publishes its own per-wheel slip angles, loads and forces as ground truth. Transplanted unchanged from a small-scale platform, it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients on their box constraints: not a solver, bounds or tuning failure but a loss of excitation created by the pipeline's own virtual-data step. The friction the synthetic rollout can demand, $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$, is a property of the rollout configuration and not of the tire. A grid over rollout speed, sweep amplitude and true peak factor with a known tire confirms it: below the tire's own peak the fit returns the excitation rather than the tire, whatever the solver and start point. We then correct the pipeline, taking rollout speed from the measured lap, bounding the sweep to the observed steering range and freezing the regularisation target, and gate the model-to-controller boundary on derived quantities instead of per-coefficient bounds. Over three timed laps no identified coefficient rests on a bound, and the corrected pipeline holds a peak factor within 0.05 of the plant's effective friction, against 0.77 for the inherited configuration, while halving the lateral tracking error under MPC. All results are in simulation, one run per configuration.**

***Index Terms*—Dynamics, Calibration and Identification, Autonomous Vehicle Navigation, Wheeled Robots, Simulation and Animation.**

## I. Introduction

A model-based racing controller is only as good as the vehicle model it carries [1], and in the lateral channel that model is dominated by the tire, the physical bound on manoeuvrability and the least accessible component to measure [2]. Classical characterisation takes the vehicle off the track: steady-state manoeuvres give clean data, at a logistical cost a race weekend rarely affords [3]. Identifying on track removes that cost and brings back transients, noise and, more awkwardly, a racing line that is a poor excitation signal for a nonlinear model.

Dikici *et al.* [3] propose the hybrid remedy that is the starting point of this paper. A small residual network is trained on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model. The corrected model is then rolled out through a *synthetic*, quasi-steady steering sweep, and Pacejka coefficients are fitted to that sweep. Because the sweep is generated rather than driven it can be made steady and noise-free, and because the network only has to interpolate within its training distribution, its generalisation weakness stays contained. The procedure iterates, each fit becoming the next nominal model, and on a 1:10 platform it reports a 3.3× lower one-step RMSE than nonlinear least squares under noise.

This paper asks what the pipeline identifies when it moves from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16° of road-wheel lock, on a reference profile of 9.6–25.8 m/s. The question is identifiability, not parameter transfer, and two scale-dependent mechanisms, both latent at 1:10, break the estimate. The first is a collapse of excitation inside the virtual-data step: the Pacejka fit never sees the recorded lap, only the synthetic rollout, whose reachable friction $\mu_{\text{reach}} \approx v^2\delta_{\text{sweep}}/(gL)$ is a property of the rollout configuration and not of the tire. A rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire's peak the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$, only $BCD$ is identifiable, and the bounded optimiser resolves the remaining freedom on a box edge [4]. The second is that lateral demand scales as $v^2$, so a bad grip estimate stays silent until it is catastrophic: one identified set with a peak friction coefficient of 0.40 tracked at 0.16 m of cross-track error at 11.1 m/s and 41.89 m at 27 m/s (Section VII). A scaled platform never reaches the speed at which that error becomes observable.

One difference works in our favour. Because the study runs in the CARLA simulator [5] through a purpose-built ROS 2 interface, the simulator's per-wheel telemetry (slip angle, normal load, lateral force) is available as ground truth, so an on-track identifier can be scored against the forces the plant actually generated. We contribute an identifiability analysis of the virtual-data step (Section V-B) giving the bound above and an excitation-aware rollout, verified in Section VI-B on a grid over rollout speed, sweep amplitude and true peak factor with a known tire. From it follow corrections on both sides of the fit: an extrapolation-bounded sweep, a frozen regularisation target, a measurement chain built on the achieved road-wheel angle, measured mass and geometry, and a friction warm start applied only as a floor (Sections V-C–V-E). We also gate admissibility on derived physical quantities rather than on per-coefficient bounds (Section VII), and evaluate in closed loop against the plant's own forces, where the corrected pipeline preserves a peak factor within 0.05 of the plant's effective friction while the inherited configuration rails on its box constraint, and halves the lateral tracking error under MPC. All results are in simulation, one run per configuration.

## II. Related Work and Position

The Magic Formula [2] is the standard empirical description of steady-state tire force generation, and it is the model identified here. Classical characterisation obtains its coefficients from steady-state manoeuvres in a controlled space: clean data at high logistical cost [3]. On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline. The recursive alternative, a filter carrying states and parameters together, is subject to the same excitation condition, and expresses a violation as a parameter covariance that never shrinks rather than as a coefficient sitting on a bound. Among friction estimators, Hsu *et al.* [6] is the closest prior art to our warm start. They recover the friction limit below the peak from the *curvature* of the force response, read through the pneumatic trail in steering torque. We read the same curvature on the axle force curve reconstructed from IMU and odometry, with no steering-torque sensor. Gaussian-process residuals are accurate, but they carry a continuing computational cost and are usually evaluated starting from an already accurate physical model [1]. On the physics-constraint axis the closest work is Deep Dynamics [7], which estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer that clamps them into nominal ranges. Our gates share the motivation but sit elsewhere, and we report as a finding that per-coefficient box constraints are *insufficient*. The degenerate fit that motivated this work satisfied every box constraint with five of eight coefficients exactly on a bound. A later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 14.5 m/s. Wu *et al.* [8] extend the same baseline concurrently and independently, with a vision-based friction prior, an S4 residual and Nelder–Mead extraction. Our pipeline independently contains an S4D residual [9], a friction warm start and the same solver options, so we claim none of the three as contributions. Neither prior work is the fix: the degeneracy is located in the rollout excitation, not in the initialisation, and to our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not been stated before.

CARLA [5] supplies the simulation infrastructure. Its native dynamics come from the PhysX wheeled-vehicle model [10], a saturating friction-and-stiffness law rather than a Magic Formula fit to measured data, so this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth coefficients.

## III. Identification Problem

The identification model is the dynamic single-track model of the baseline, with mass $m$, yaw inertia $I_z$ and centre-of-gravity distances $l_f$, $l_r$, $L = l_f + l_r$; longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \tfrac{1}{m}\big(F_{y,r} + F_{y,f}\cos\delta - m v_x \omega\big), \tag{1}$$

$$\dot{\omega} = \tfrac{1}{I_z}\big(F_{y,f} l_f \cos\delta - F_{y,r} l_r\big), \tag{2}$$

with slip angles

$$\alpha_f = \delta - \arctan\!\Big(\tfrac{v_y + l_f\omega}{v_x}\Big), \quad \alpha_r = -\arctan\!\Big(\tfrac{v_y - l_r\omega}{v_x}\Big), \tag{3}$$

and axle forces from the lateral Magic Formula

$$F_y = F_z\, D \sin\!\big(C \arctan\!\big(B\alpha - E(B\alpha - \arctan B\alpha)\big)\big), \tag{4}$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$ is discretised by explicit Euler at $T_s = 0.035$ s. The objective is to recover $\Phi_p$ from a single 30 s segment of ordinary lap data, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities: the linearised axle cornering stiffness $C_i = F_{z,i} B_i C_i^{\mathrm{MF}} D_i$, the slope of (4) at zero slip (writing $C^{\mathrm{MF}}$ for the shape factor to avoid a clash), and the implied grip ceiling

$$a_y^{\max} = \big(D_f F_{z,f} + D_r F_{z,r}\big)/m. \tag{5}$$

The linear model is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\mathrm{crit}} = L\sqrt{C_f C_r / (m(l_f C_f - l_r C_r))}$.

### A. Practical identifiability under limited excitation

Two properties of (4) are used throughout. Its slope at zero slip is $F_z BCD$, so the cornering stiffness depends on the coefficients only through that product, and $E$ drops out. If the data never approaches the peak, (4) linearises to $F_y \approx F_z(BCD)\alpha$ and $B$, $C$, $D$ become mutually unidentifiable. Given data that does reach the peak, (4) is *structurally* identifiable. What fails here is *practical* identifiability [11]: under a specific excitation the likelihood is flat along the $BCD$ direction. This is the standard persistent-excitation condition [4] written on this model, and we claim neither the structural result nor the excitation condition as novel. Our claim, demonstrated in Section VI-B, is that the baseline's virtual-data step can silently violate that condition, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge. The violation surfaces as a converged fit rather than as a failure.

### B. Baseline pipeline

The baseline repeats four stages [3]. A network is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$. The corrected model is integrated open-loop at constant speed while the steering is ramped from 0 to $\delta_{\mathrm{sweep}}$, which produces synthetic quasi-steady data. Assuming $\dot{v}_y = \dot{\omega} = 0$ there, axle forces follow from the yaw balance,

$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \qquad F_{y,f} = \frac{m l_r v_x \omega}{L\cos\delta}, \tag{6}$$

slip angles from (3), and $\Phi_p$ is fitted by bounded least squares. That fitted $\Phi_p$ becomes the next iteration's nominal model. The point that drives the rest of this paper is that the fit *never sees the recorded lap*. It sees the rollout. The lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

## IV. Simulation Architecture and Data

The pipeline runs against a Formula Student vehicle model in CARLA [5] through a C++ ROS 2 bridge that drives the vehicle and republishes its state. Two properties of that architecture make the identifiability claims checkable. CARLA's PhysX solver computes each wheel's slip angle, normal load and lateral force internally, and the bridge republishes them on the same clock as the odometry the identifier consumes, so an identified tire model is scored against the forces the plant generated rather than against a second model, and that channel never enters the identification loop. The server also runs synchronously at a fixed 0.033 s step, with every sample stamped from the simulation clock, so the one-step residual target is well defined.

The plant is the driverless Formula Student–class vehicle on the `silverstone` map (CARLA 0.9.16), whose lateral forces come from the PhysX tire model [10] under a configured friction coefficient and lateral stiffness, not from a Magic Formula, so the simulator has no "true" Pacejka coefficients to recover and the forces PhysX computes at each wheel are the ground truth instead. At zero longitudinal slip and zero camber that model collapses to

$$F_y(\alpha) = \operatorname{sgn}(\alpha)\,\mu F_z\,S_1(K), \qquad K = \frac{C_{\mathrm{lat}}|\tan\alpha|}{\mu F_z}, \quad (7)$$

with $C_{\mathrm{lat}} = F_z^{\mathrm{rest}} k_y S_1(3\bar{F}_z/k_x)$, $\bar{F}_z$ the normalised load and $S_1(K) = \min(1, K - K^2/3 + K^3/27)$ the engine's smoothing function. It saturates at $\mu F_z$, first reached at $K = 3$, and never falls off past that peak, so a Magic Formula can match it up to the peak and no further. Its scale is set by $\mu$ and $C_{\mathrm{lat}}$, both published by the simulator, so the plant's curve is read off rather than fitted: $\mu = 1.05$ effective (a configured wheel friction of 1.5 times the road surface's 0.70, with a realised peak axle friction of 1.00–1.05), $k_y = 17\,$/rad, $k_x = 2.0$, giving axle slopes of 20.4 and 19.0 kN/rad front and rear at $\bar{F}_z = 1$ and the static axle loads $F_{z,f} = 1373\,$N, $F_{z,r} = 1274\,$N, and a peak first reached at $12.0^\circ$ of slip. On 1804 ground-truth wheel samples from a live lap, each predicted from its own measured $\alpha$, $F_z$ and $\mu$, (7) returns $R^2 = 0.99$ at 16.1 N RMSE front and 14.7 N rear. The vehicle itself measures $m = 269.6\,$kg (total static wheel load 2644.5 N over $g$, not the configured body mass of 240.0 kg), $L = 1.5334\,$m with $l_f/l_r = 0.7381/0.7953\,$m, an assumed $I_z = 51.1\,\mathrm{kg\,m}^2$, and $16.0^\circ$ of road-wheel lock.

A model-free pure-pursuit controller drives the vehicle around the reference raceline. It needs no tire knowledge beyond the wheelbase and is therefore a valid bootstrap, as in the baseline [3]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer for 30 s, gated on $v_x > 1\,$m/s, matching the baseline's data budget, and re-identification retriggers every 30 s. The model is discretised at $T_s = 0.035\,$s, a setting carried because the identification proved insensitive to it: identifying at 0.020 s returns the same last accepted coefficient set to four decimal places in the corrected configuration, and still leaves $D$ on its 2.0 bound in the inherited one, so the box-edge signature below is not an artefact of the step mismatch. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ at unchanged $v_x$.

## V. Identification Methodology

The pipeline keeps the four-stage loop of Section III-B. What changes is the configuration of the virtual-data step (Sections V-B–V-C), the measurement chain that feeds it (Section V-E), and the contract at its output (Section VII): a coefficient resting on a bound is reported as a degeneracy, a set failing the gates is returned as UNIDENTIFIED, and the previous model is kept until the controller acknowledges a new one.

### A. Residual model

The default residual is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, 58 parameters, trained full-batch on an MSE objective. Two options shape it with vehicle physics rather than with capacity. The first augments its inputs with the axle slip angles of (3), since (4) depends on $(\alpha, F_z)$ and nothing else; the second adds to the MSE [7] a quasi-steady consistency term, the odd symmetry $e(-\mathbf{z}) = -e(\mathbf{z})$ of a symmetric vehicle, and a soft Lipschitz bound. A third replaces the memoryless MLP with an S4D state-space residual [9] over a window of $W = 20$ samples; we do not claim it as a contribution [8], and the identifiability result below holds for either residual.

### B. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-B: a constant-speed rollout at $v$ with the steering ramped to $\delta_{\mathrm{sweep}}$. In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2\kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can *demand* of the tire is

$$\mu_{\mathrm{reach}} \;\approx\; \frac{v^2\,\delta_{\mathrm{sweep}}}{g\,L}. \quad (8)$$

Equation (8) contains no tire parameter: it is a property of the rollout configuration alone, and an *upper* bound rather than an equality, since $\kappa \approx \delta/L$ is the Ackermann curvature while the steady-state curvature a compliant vehicle holds is smaller. Reading it as a ceiling is all the argument needs. If even the ceiling sits below $D$, the peak is never reached, the synthetic data stays on the linear ramp of (4), only $BCD$ is identifiable, and the bounded optimiser resolves the residual freedom on a box edge.

The baseline selects the rollout speed as $v = \mathrm{clip}(\overline{v_x}, 2, 4)\,$m/s, representative of a 1:10 platform. On our vehicle, with $\delta_{\mathrm{sweep}} = 0.4\,$rad and $L = 1.5334\,$m, (8) gives $\mu_{\mathrm{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than $0.43\,g$, whatever the tire is. At 1:10 scale, with

$L \approx 0.32\,\mathrm{m}$, the same speed yields $\mu_{\mathrm{reach}} \approx 2.0$ and the pathology is invisible.

The rollout speed is therefore taken from the speed actually driven, clamped below at 15 m/s, and the pipeline warns whenever (8) falls below the prior's $D$, which turns a silent degeneracy into a startup diagnostic. The same bound fixes the data-collection speed: data should be gathered at $v \geq \sqrt{gLD/\delta_{\mathrm{sweep}}}$.

### C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure: the rollout queries the residual at steering angles it has never seen, the observed $|\delta|$ on a pure-pursuit lap staying below $\approx 0.06\,\mathrm{rad}$ while the sweep ran to $0.4\,\mathrm{rad}$. The fit reads that extrapolation as additional grip, and because each iteration's answer became both the next nominal model *and* its regularisation target, the error compounds: $D$ walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed $1.7\,g$ on a vehicle measuring $1.0\,g$. Two changes close the loop.

*Extrapolation bound.* The sweep terminates at $\delta_{\mathrm{sweep}} = \mathrm{clip}(\lambda_e P_{99}(|\delta_{\mathrm{obs}}|), \delta_{\min}, \delta_{\max})$ with $\lambda_e = 1.5$, $\delta_{\min} = 0.05\,\mathrm{rad}$, $\delta_{\max} = 0.54\,\mathrm{rad}$, so the network is never queried more than 50 % beyond the steering range it was trained on. That $\delta_{\max}$ lies above the plant's $16.0^\circ$ mechanical lock is deliberate: the sweep is a *virtual* rollout, and clamping it to the lock would cap $\mu_{\mathrm{reach}}$ a second time, by the actuator instead of by the rollout speed. In practice the binding limit is the data-driven term.

*Frozen prior.* The Tikhonov target is frozen at the configured prior $\Phi_p^0$ for all iterations, while the optimiser's *start point* still tracks the previous iteration. The per-axle objective is

$$\min_{\phi} \sum_k \rho\big(r_k(\phi)\big) \; + \; \sum_j w_j \big(\phi_j - \phi_j^0\big)^2, \qquad (9)$$

with $r_k$ the force residual, $\rho$ a soft-$\ell_1$ robust loss and $w_j$ scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists for one purpose: to hold the coefficients the manoeuvre does not excite, $E$ above all, at their nominal value instead of on a box edge.

### D. Pacejka fit

Fitting applies (6) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle. Relative to the baseline we raised the jittered multi-starts from 1 to 8 and added a simplex and a global differential-evolution solver [12] beside trust-region reflective [13]. Neither addresses the degeneracy: at a 4 m/s rollout every tire tested fits back as $\hat{D} \approx 0.5$ with coefficients railed, and the answer moves with neither the seed nor a jitter seven times wider.

### E. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator. First, $\alpha_f$ in (3) was built from the steering setpoint. The achieved/commanded ratio is 0.76, so $\alpha_f$ came out $\approx 25\,\%$ too large and the fitted front stiffness $\approx 20\,\%$ too small: 13725 N/rad against the 17198 N/rad fitted to the simulator's per-wheel forces over the same run. The achieved angle recovers 18486 N/rad. Second, $m$ and the wheelbase come from measured static wheel loads rather than from the configuration file, whose 240.0 kg against the 269.6 kg the wheels carry biased the friction estimate 11 % low on its own.

Third, the peak factor gets a prior, which the inherited pipeline did not have. Newton–Euler on (1)–(2), solved for the forces, gives $F_{y,f} = (I_z\dot{\omega} + l_r m a_y)/(L\cos\delta)$ and $F_{y,r} = (l_f m a_y - I_z\dot{\omega})/L$ from the IMU's lateral acceleration and the yaw acceleration, with no tire model at all; stiffness and peak friction are then fitted per axle to the two-parameter Fiala brush model [14], whose parameters the manoeuvre can support where the four of (4) cannot. What lets this beat the utilisation bound is that $\mu$ enters the brush model through the *curvature* of $F_y(\alpha)$ and not only through its asymptote, so a 30 % departure from the force's own linear extrapolation already pins it, well below the limit. That also states the failure mode, the estimate being unbounded in the limit of a linear curve, so the value is released only when grip utilisation $\max|F_y|/(\hat{\mu}F_z) \geq 0.40$, the level the lateral-dynamics friction literature converges on [15], and relative precision $\sigma_{\hat{\mu}}/\hat{\mu} \leq 0.15$ both hold, and $\hat{\mu}$ does not rest on its search bound; otherwise the module reports the utilisation bound. The utilisation test uses the *fitted* $\hat{\mu}$ and is on its own circular. The two companion conditions break that circularity, and the last rule makes whatever still slips through one-sided: the released value may only ever *raise* $D_f, D_r$. That floor matters because the frozen Tikhonov target of (9) is taken immediately after the warm start, and because the controller re-ingests its reference speed profile from $a_y^{\max}$ on each accepted update, so a lowered $D$ would rewrite the reference under the running controller. For the same reason the warm start runs on the first identification cycle only.

### F. Validation protocol

Three levels are used. One-step prediction on a held-out run admits samples only above 1 m/s, as the identification data does. Direct force comparison against the per-wheel telemetry separates the error of the fit from that of representing a PhysX tire as (4). Closed-loop tracking under the MPC is split by active controller, so the acquisition phase is not scored against a model it does not use. All three score the *accepted* model only: a rejected submission is a model the car never drove on.

## VI. Experiments and Results

### A. Setup

Every run uses a *static* friction surface: the effective coefficient the identifier is scored against is 1.05, constant to the last digit the simulator reports over all 8186 and 9355 validation samples. Both runs follow the same minimum-curvature raceline on a recorded centreline of the CARLA silverstone circuit: a closed 5828.2 m lap, reference speed 9.62–25.78 m/s, minimum corner radius 18.7 m, peak lateral demand $0.51\,g$, roughly half the $1.0\,g$ the plant sustains. It replaces an earlier trajectory whose $1.22\,g$ demand the gate of Section VII rejected.

Two configurations of the pipeline are run back to back on that trajectory, three timed laps each, after an untimed out-lap to the raceline's $s = 0$. They differ in four settings and nothing else: residual architecture (MLP against S4D), objective (one-step MSE against the physics-informed loss), Pacejka solver (trust region against differential evolution), and friction warm start (off against on). `Baseline-NN-MSE` is the inherited configuration evaluated at full scale on the corrected rollout, so the comparison isolates those four settings rather than reproducing the published pipeline in full. Because they move together it is a configuration comparison and not a per- factor ablation, and since the excitation-aware rollout is active in *both* arms it is not a test of this paper's central contribution, which is evidenced separately in Section VI-B.

The prior $\Phi_p^0$ both arms start from, and that (9) pulls toward, was obtained by fitting (4) to the simulator's per-wheel telemetry over a dedicated limit-excitation run reaching $\alpha = 0.37\,\text{rad}$ and $1.50\,g$. Two consequences need stating plainly: that is the same channel the force and friction results are scored against, so the prior is not independent of the validation data, and the manoeuvre that produced it is one the pipeline is not allowed to run, since the raceline peaks at $0.51\,g$. On all six accepted identifications the `Ours` arm returns the eight prior coefficients to four decimal places, so the peak-friction accuracy below is that of a prior the fit did not move, not of a peak factor recovered from the lap. What the corrected pipeline demonstrably does is *preserve* a physically valid peak factor where the inherited configuration destroys it by railing $D$ on its bound.

### *B. Identifiability of the peak factor against rollout excitation*

The claim of Section V-B is about the rollout configuration alone, so it is tested where the answer is known exactly rather than in CARLA. A tire with a known peak factor $D$ is rolled out through the synthetic sweep of stage 2, the shipped rollout with the residual network set to zero, and fitted back with the shipped solver, *unregularised* and started from the true coefficients, so everything the fit loses is lost to excitation and not to the prior, the start point or a local minimum. Rollout speed $v$, sweep amplitude $\delta_{\text{sweep}}$ and true $D$ are swept, each cell repeated over five noise realisations at the measured one-step persistence RMSE of the two CARLA runs ($\sigma_{v_y} = 0.009\,\text{m/s}$, $\sigma_\omega = 0.005\,\text{rad/s}$). The configuration axes are swept one at a time, since $\mu_{\text{reach}}$ depends on both and the claim that it, not the speed, governs the fit cannot be tested by moving the speed alone. Repetition is over noise rather than over the solver because the noise-free fit is seed-invariant here: five multi-start seeds and a jitter seven times wider return the same coefficients to four decimals.

Figure 1 says three things. First, the inherited 4 m/s rollout is uninformative about $D$ at *every* grip level tested: a $0.70$, a $1.05$, a $1.40$ and a $1.80$ tire all come back as $\hat{D}$ between 0.50 and 0.59, the fit reporting $\mu_{\text{reach}}$ and not the tire, with two to seven of the eight coefficients on a bound. Second, the boundary is $\mu_{\text{reach}} \gtrsim D$ and not a speed: at 6 m/s ($\mu_{\text{reach}} = 0.96$) the $0.70$ and $1.05$ tires come back within 3 % while the $1.40$ and $1.80$ tires are 13 % and 30 % short, and above $\mu_{\text{reach}} = 1.7$ every tire is within 4 % with no systematic railing. That boundary is soft rather than sharp, the $1.05$ tire being recovered slightly *below* its own $D$ because a rollout stopping just short of the peak still traverses the curvature separating $B$, $C$ and $D$, so the bound predicts where the information runs out. Third, the failure is not the solver's: the noise-free fit is invariant to the multi-start seed and the jitter width, so no solver or start point recovers a coefficient the data does not contain.

Panel (b) also shows that the failure is not a common bias. An under-speeded rollout returns *too little* grip; an under-swept one returns too much. At $\delta_{\text{sweep}} = 0.06\,\text{rad}$ the 1.05 tire comes back as 1.30 and the 1.80 tire rails on its bound. What the two axes share is that $\hat{D}$ stops being a function of the tire, the direction of the error being set by the rollout: the signature of an unidentified parameter rather than a biased one. They do *not* collapse onto a common $\mu_{\text{reach}}$, since (8) uses the Ackermann curvature and tracks the realised demand only while understeer is small. Against the realised $\max|v_x\omega|/g$ (panel c) both behave alike. Equation (8) is therefore a cheap screening bound, conclusive when it falls below $D$, not a measure of the excitation achieved. With the residual set to zero the structure is exactly matched, so this is a best case and the real pipeline degrades at least this much.

### *C. Identified tire model and peak friction*

Figure 2 is the cross-run comparison, and its bottom-right panel puts both identified curves against the plant's own curve (7). No coefficient of `Ours` rests on a box constraint in any of its six accepted identifications. The last accepted set is $B, C, D, E = 7.076, 1.346, 1.009, -2.000$ front and $7.873, 1.383, 1.002, -1.024$ rear, inside $B \in [4, 20]$, $C \in [1.2, 2.2]$, $D \in [0.4, 2.0]$, $E \in [-3, 1]$. Its front $E = -2.000$ is the configured prior, not a bound, and is exactly the coefficient the frozen Tikhonov target of (9) holds at its nominal value where the lap gives no support, while the unregularised fit moves it to $-2.264$ on no evidence. `Baseline-NN-MSE` puts $D$ on its upper bound of 2.000 in the first and the last of its six identifications and wanders over $D_f \in [1.241, 2.000]$ in between. A peak factor of 2.000 against a plant realising 1.05 is less a 90 % parameter error than the box-edge signature of Section III-A: the coefficient is not identified, and no per-coefficient bounds check distinguishes that from a fit. With the surface static, `Ours` returns $D_f = 1.009$ and $D_r = 1.002$ in *every* one of its six identifications, so its $\mu$ error is a constant 0.041/0.048 against 0.762/0.772, 16–19× worse. The warm start's floor binds on no cycle here.

The price `Ours` pays is in the linear range (slopes at $\bar{F}_z = 1$ and the same static axle loads throughout): 13.2 kN/rad front and 13.9 kN/rad rear against the plant's 20.4 and 19.0, 35 % and 27 % low, where `Baseline-NN-MSE` gives 19.1 and 18.1, 6.5 % and 4.6 % low. The understeer sign is correct in both, so both pass the stability gate, but the baseline's margin is 2 % ($l_f C_f = 14.10\,\text{kN m/rad}$ against $l_r C_r =$

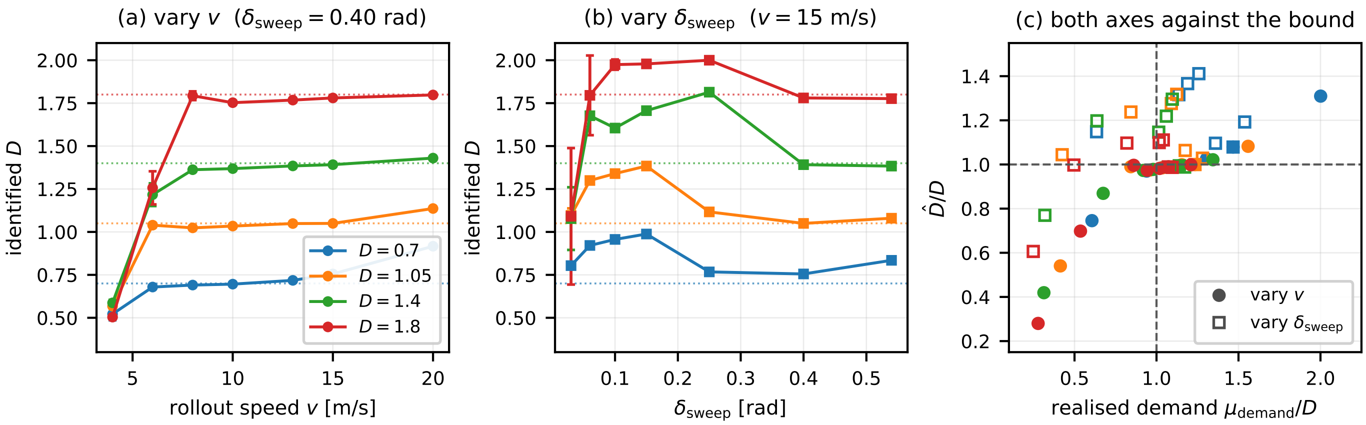


Fig. 1. Peak factor $\hat{D}$ recovered from a known tire, unregularised and started from the true coefficients; one series per true $D$ (dotted), mean $\pm$ the 95 % interval over five noise realisations. (a) and (b) sweep the two arguments of (8) one at a time, the inherited configuration being the leftmost point of (a); starving $\delta_{\text{sweep}}$ at a fixed 15 m/s destroys $D$ as thoroughly as starving $v$, so neither axis is a speed effect. (c) reads both against the demand the rollout *realises*, normalised by $D$: left of the dashed line $\hat{D}$ reports the excitation, low where speed limits it (filled), high where sweep amplitude does (open).

14.38 kN m/rad) where `Ours` keeps 12 %: a set that lands 2 % from an oversteering pair on a coefficient it did not identify is the case the gate exists for. Only `Baseline-NN-MSE`'s grip ceiling is a claim the plant cannot honour, (5) giving 2.00 $g$ against 1.01 $g$ for `Ours` and $\approx 1.0\,g$ measured. Both effects coexist because the lap covers only $P_{99}(|\alpha|) = 2.2^\circ$ against the $12.0^\circ$ at which the plant's curve peaks: it constrains the slope and says almost nothing about the peak.

### *D. Force, prediction and closed-loop tracking*

The identified curve, scored at the measured $(\alpha, F_z)$ against the plant's per-wheel force, orders the arms differently across axles. `Ours` is better at the rear (40.8 N RMSE, $R^2 = 0.979$) and worse at the front (70.7 N, $R^2 = 0.944$, against 45.9 N and 0.978), and the inherited configuration's lower front RMSE is bought with a maximum absolute error 1.6$\times$ heavier at the front and 2.6$\times$ at the rear. That is the linear-range deficit showing up as force error, and it is the one axis on which the inherited configuration is better: at $2.2^\circ$ of slip both curves are far below their peaks, so a railed peak factor costs nothing while a 35 % slope deficit costs 71 N of front-axle RMSE. What separates the two models is not the force at the slip angles the lap visits, but the grip ceiling those forces never probe. Neither model beats persistence at the 0.035 s model step ($v_y$ RMSE 0.027 against 0.022, $\omega$ 0.032 against 0.068, persistence 0.009 and 0.005). A one-step metric on a smooth trajectory rewards not moving, so it is not evidence for either tire model.

Both runs finish three timed laps with zero emergency halts and the same state sequence (bootstrap pure pursuit, running pure pursuit, handover, running MPC), with no fallback after handover. Under MPC, where the identified model enters the loop, `Ours` tracks at 0.092 m RMS lateral error against 0.177 m, a 48 % reduction, with maximum 0.490 m against 0.696 m. ITAE is neither scale-free nor comparable across runs of different length, so we compute two normalised forms from the same samples: IAE$/T$ is 0.085 m against 0.132 m and ITAE$/(T^2/2)$ is 0.059 m against 0.111 m, and all three order the runs alike. The split by controller puts the whole difference in the MPC phase: the pure-pursuit phases give ITAE 211 and 122, ordered the other way and two orders of magnitude smaller. The QP cost is unchanged, both solve times being two orders of magnitude below the 50 ms control period with the orderings disagreeing, so the coefficients change the model the QP carries, not its size.

### *E. Friction warm start and identification cost*

The friction warm start is evaluated on a synthetic brush-tire plant rather than in CARLA, because only a synthetic plant supplies a $\mu$ known exactly and an excitation that can be swept independently of it. At 45 % grip utilisation, a level an ordinary lap reaches, the utilisation bound returns 0.473 against a true $\mu = 1.05$ while the brush fit returns 1.048. It returns that same 1.048 at 63 %, 80 % and 100 %: flat in excitation, where the bound tracks only the grip used. Across true $\mu \in \{0.7, 1.05, 1.4\}$ the fit stays within 5 %, and 12 % with odometry and IMU noise; below the 40 % floor the gate suppresses it and reports the bound, so the failure is visible rather than silent. One identification, six iterations on a 30 s lap, takes 20.9 s, inside the re-identification period.

## VII. Integration with the Controller

The identified model leaves the pipeline through one interface. A nonlinear MPC path tracker (`acados`, horizon $N = 20$ at 20 Hz) carries the same single-track model as the identifier and consumes $\Phi_p$ through (4). Nothing else about it changes when a new identification arrives. A coefficient set can lie strictly inside every box constraint and still be unusable, in three distinct ways we observed, each needing a gate on a *derived* quantity rather than on a coefficient.

*Grip ceiling versus trajectory demand.* The controller computes $\max_k(v_{x,k}^2|\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (5) falls below it. On the earlier trajectory a degenerate fit reported a 0.40 $g$ ceiling against a 1.22 $g$ demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a plausible set

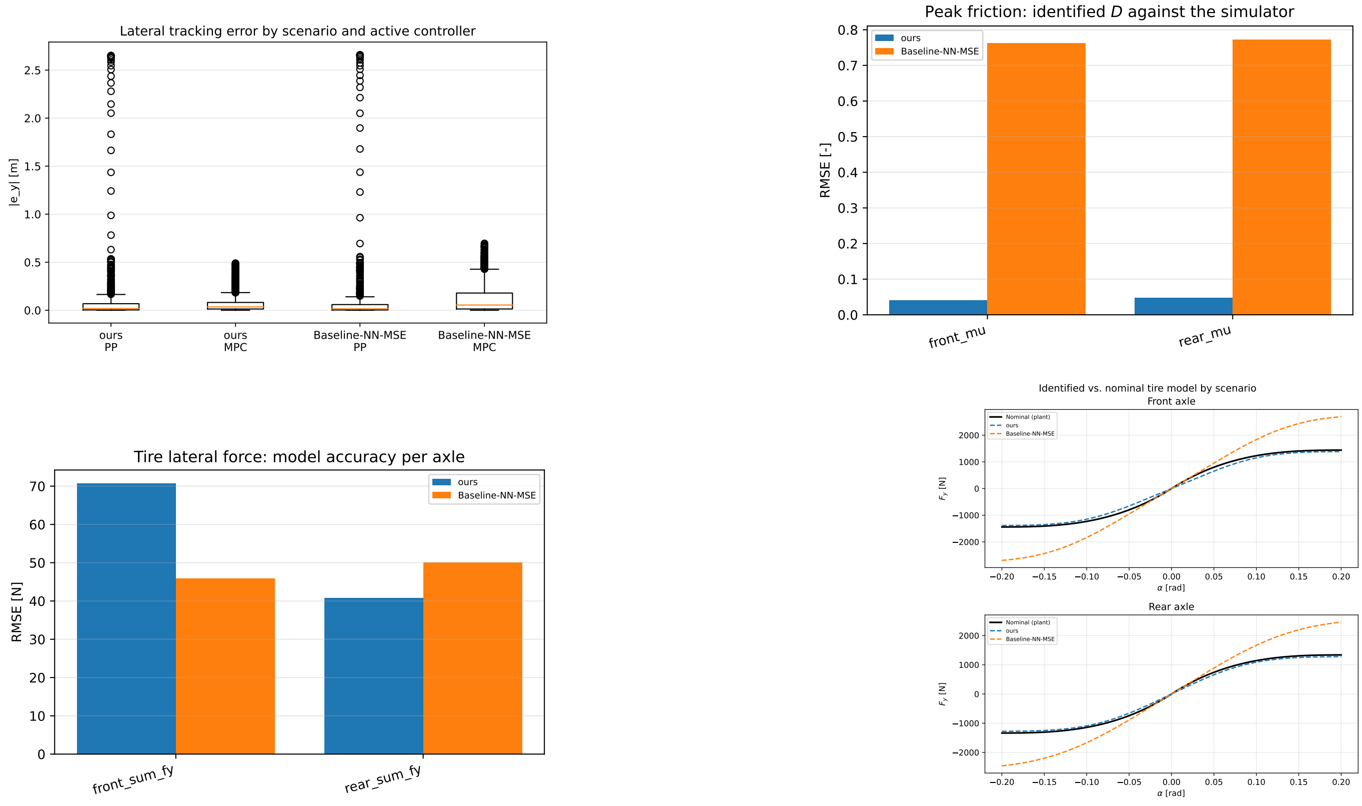


Fig. 2. Cross-run comparison, every panel scored against the simulator's own telemetry: lateral tracking error by active controller (the pure-pursuit boxes precede handover and do not use the identified model); peak-factor error against the plant's $\mu = 1.05$; axle force RMSE, the one family the inherited configuration wins, and only at the front; and the identified tire curves against the plant's own curve (7) at each axle's static load (black), where `Baseline-NN-MSE`'s railed peak factor carries roughly twice the plant's force at the edge of the range while `Ours` tracks the peak and undercuts the slope. **Single run per configuration: no bar or box carries a confidence interval (Section VIII-A).**

($D = 1.5$) at the same speed through the same controller, figures from an offline harness quoted as the observation that motivated the gate. The gate is one-sided and catches only a ceiling that is too low, so `Baseline-NN-MSE`'s $2.00\,g$ passes it.

*Understeer sign and critical speed.* A set with $l_f C_f > l_r C_r$ whose $v_{\text{crit}}$ falls inside the controller's speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to $\rho^N$ in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 14.5$ m/s at the measured geometry and mass, inside the controller's range on every trajectory here.

*Integration stiffness.* Even a stable set can be stiff relative to the control period: one identified set gave $\rho(A_d) = 547$ at 5 m/s for a single explicit RK4 step at $dt = 0.05$ s, so the model sub-steps adaptively from its own fastest lateral mode ($|\lambda| h \leq 1.5$), restoring $\rho \approx 1.000$ at a QP median of 0.36 ms.

The gates are the contract between identification and control. A passing identification is a model the controller can be held responsible for; a failing one is reported as unidentified rather than deployed. Two controller-side changes were also needed before any identified model could be evaluated at full-scale speeds. Reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve. We record both for completeness, not as identification results.

## VIII. Discussion and Limitations

The reachable-friction bound (8) is kinematic, carrying no tire parameter, no simulator-specific quantity and no property of the residual, so it applies to any implementation of the virtual-steady-state-data strategy of [3], in simulation or on hardware. Its practical content is a design rule, *the rollout must demand at least the friction one intends to identify*, and where it cannot, $D$ should be reported as unidentified rather than fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be logged by any bounded tire fit, being otherwise indistinguishable from a successful one. So is the insufficiency of per-coefficient box constraints for admissibility: both failures we encountered, a degenerate set and a pairwise oversteering one unstable inside the controller's range, satisfied every box constraint, and both are properties of the parameterisation, not of CARLA.

What does not generalise is every number attached to this plant. The tire values describe a PhysX model with a configured friction coefficient, so agreement with them shows only that the identifier recovers the plant it observes; hardware replication is necessary before any accuracy claim on a vehicle. The gap between configured wheel friction

(1.5) and realised peak axle friction (1.00–1.05), and the 0.76 achieved/commanded steering ratio, are properties of this configuration. The lessons generalise and the magnitudes do not: score against the effective friction the simulator publishes, identify against the achieved actuator state.

### A. Limitations

*One run per configuration, so nothing in Fig. 2 carries an uncertainty.* The 0.092 m against 0.177 m RMS tracking difference has no confidence interval, and no evidence that it exceeds run-to-run variation, which we did not measure; nothing prevents repetition, so this is a gap in the evidence, not the method. *Single surface, vehicle and trajectory.* Friction is static at 1.05, and we do not run the surface-change adaptation experiment, arguably the strongest argument for on-track identification. *The friction estimator's excitation sweep is synthetic.* On the CARLA runs the warm start sees only the excitation the lap supplies. *The two configurations differ in four settings at once.* Section VI therefore attributes nothing to the residual, the loss, the solver or the warm start individually, and with the excitation-aware rollout active in *both* arms the closed-loop comparison is not evidence for the identifiability correction. The sweep of Section VI-B is. *Longitudinal dynamics are out of scope,* as in the baseline.

## IX. Conclusion and Future Work

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [3] at full vehicle scale, in a CARLA architecture exposing the plant's per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform it fails in a specific, diagnosable and correctable way, and not through the solver, the bounds or the tuning. The virtual-data step's own reachable friction $\mu_{\mathrm{reach}} = v^2\delta_{\mathrm{sweep}}/(gL)$ depends on the rollout configuration and not on the tire, and a rollout speed appropriate at 1:10 caps it at 0.43 at full scale, below the plant's peak, so the peak factor is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge.

The correction has six parts: an excitation-aware rollout speed, an extrapolation bound on the sweep, a frozen regularisation target, identification against the achieved road-wheel angle, measured mass and geometry, and a friction warm start read from the curvature of the axle force curve and applied only as a floor. Together they produce identifications with no railed coefficient, a peak factor of 1.009/1.002 against a plant realising 1.05, and the correct understeer sign. Over three timed laps on a static surface the pipeline *preserves* that peak factor to 0.041/0.048 RMSE where the inherited configuration starts and ends on its 2.0 box constraint, and halves the MPC's lateral tracking error, 0.177 m to 0.092 m RMS, at unchanged solver cost. Preserves, not identifies: this trajectory never approaches the tire's peak and the accepted fits return the configured prior to four decimals. Recovery *when the excitation reaches the peak* is shown on a known tire in Section VI-B. Each configuration is a single run, so the closed-loop margins are observations rather than estimates with a confidence interval. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and gate on derived physical quantities at that interface instead.

Equation (8) inverts into a requirement on the manoeuvre, so a controller that inserts bounded slalom excitation when the identifier reports insufficient slip coverage would move the pipeline toward active, safety-bounded input design [16]. The decisive test of every claim here is a physical vehicle: the identification code is unchanged between simulation and hardware, and only the ground truth goes away.

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